$P_m = ?$

<u>Q4</u> Any constant sequence.

$\forall n \geq 1 . x_n = a$.

It is arithmetic with constant difference 0.

It is geometric with constant ratio of 1.

<u>Ex 1.2.2</u>

$N! = N(N-1)(N-2) \ldots 3.2.1$

$$a_m = \frac{5^m}{m!}$$

Consider the ratio $\frac{a_{m+1}}{a_m}$

$$\frac{a_{m+1}}{a_m} = \frac{5^{m+1}}{(m+1)!} \cdot \frac{m!}{5^m}$$

$$= \frac{5}{m+1}$$ compare to 1 ?

$$\frac{a_{m+1}}{a_m} = \frac{5}{m+1} \begin{cases} > 1, & \text{for } m = 1, 2, 3 \\ \leq 1, & \text{for } m \geq 4. \end{cases}$$

Therefore

$$a_{m+1} > a_m \quad \text{for } m = 1, 2, 3$$

$$a_{m+1} \leq a_m \quad \text{for } m \geq 4.$$

So sequence increases at the beginning and then decreases strictly from $m = 5$

<u>Example</u>

Consider $\{x_m\}_{m=1}^{\infty}$ defined by

$$x_m = \frac{3m}{m^2+1}.$$

Determine its increasing/decreasing nature.

Considering the ratio

$$\frac{x_{m+1}}{x_m} = \frac{3(m+1)}{(m+1)^2+1} \cdot \frac{m^2+1}{3m}$$

$$= \frac{(3m+3)(m^2+1)}{(m^2+2m+2)(3m)}$$

$$= \frac{3m^3+3m^2+3m+3}{3m^3+6m^2+6m}$$

$$= \frac{(3m^3+6m^2+6m) - 3m^2 - 3m + 3}{3m^3+6m^2+6m}$$

$$= 1 - \frac{3m^2+3m-3}{3m^3+6m^2+6m}$$

< 1, since $3m^3+6m^2+6m > 0$

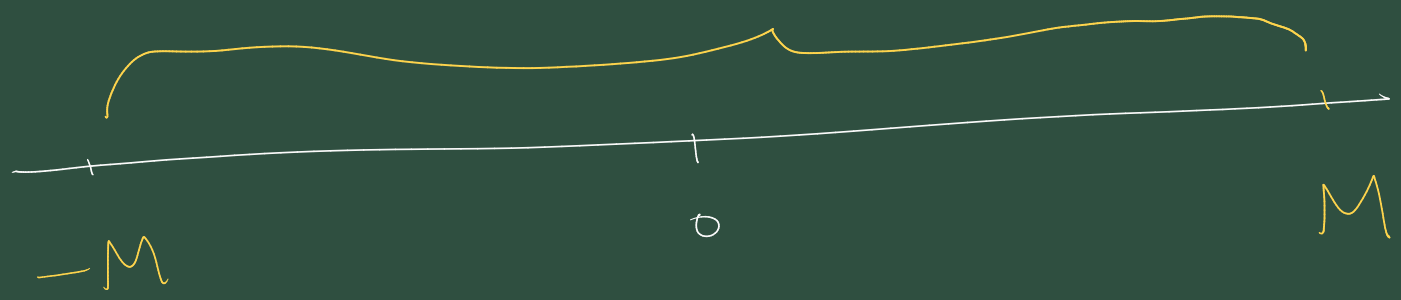
and $3m^2+3m-3 > 0$

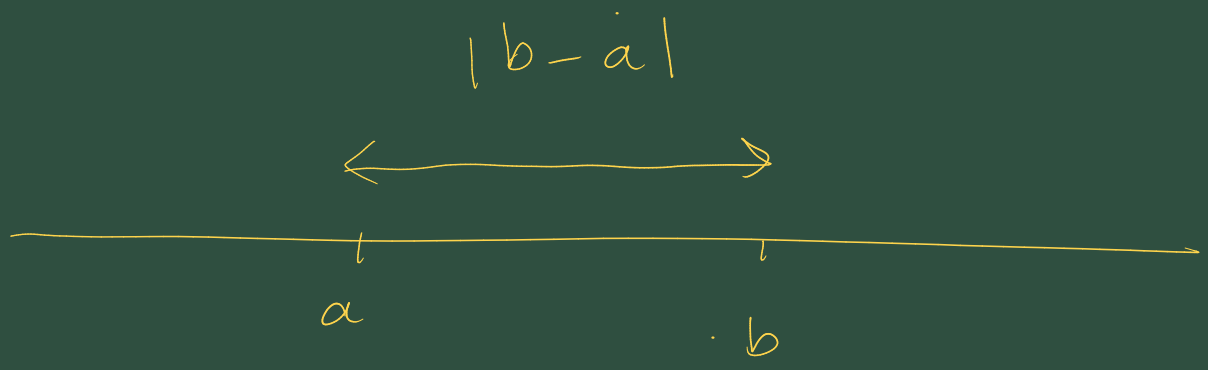
for all $m \geq 1$.

So $\frac{x_{m+1}}{x_m} < 1$

so $x_{m+1} < x_m$ for all $m \geq 1$.

So $\{x_m\}_{m=1}^{\infty}$ is strictly decreasing





$x_n \to L \iff \forall \varepsilon > 0 \ \exists N \quad n \geq N \Rightarrow |x_n - L| < \varepsilon$

Examp 1.3.6.

Let's prove $\lim_{n \to \infty} \frac{1}{n} = 0$

Proof

Let $\varepsilon > 0$ be given.

Let's examine

$$|x_n - L| = \left|\frac{1}{n} - 0\right|$$

$$= \left|\frac{1}{n}\right|$$

$$= \frac{1}{n}, \text{ as } n > 0.$$

So

$$|x_n - L| < \varepsilon$$

$$\iff \frac{1}{n} < \varepsilon$$

$$\iff n > \frac{1}{\varepsilon}$$

For $a, b > 0$

$$a < b$$

$$\iff \frac{1}{a} > \frac{1}{b}.$$

So given $\varepsilon > 0$, we can choose N to be any integer $> \frac{1}{\varepsilon}$;

then for all $n \geq N$

$$\frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

$$N > \frac{1}{\varepsilon}$$

$$\Rightarrow \frac{1}{N} < \varepsilon$$

This shows the formal definition is satisfied as

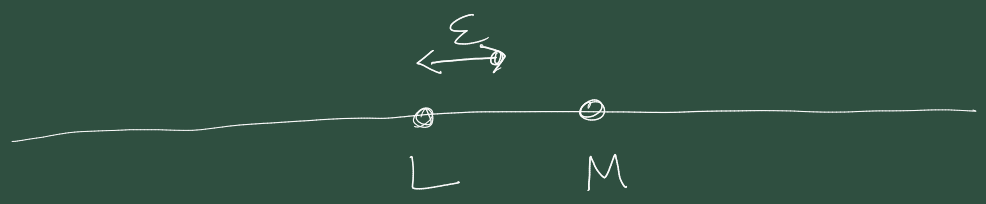
$$|x_n - L| = \frac{1}{n} < \varepsilon.$$

~~Proof~~

Theorem 1.4.2.

If $x_n \to L$ and $x_n \to M$ as $n \to \infty$
then $L = M$.

Proof. Assume (for the purposes of contradiction) that $L \neq M$.



Let $\varepsilon = \dfrac{|L - M|}{2} > 0$

So by convergence $\exists\, N_1, N_2$ such that

$\forall\, n \geq N_1 \quad |x_n - L| < \varepsilon$

and $\forall\, n \geq N_2 \quad |x_n - M| < \varepsilon$.

So now let $N = \max(N_1, N_2)$

so for $n \geq N$ $|x_n - L| < \varepsilon$ & $|x_n - M| < \varepsilon$

For $n \geq N$

$$|L - M| = |(L - x_n) + (x_n - M)|$$

$$\leq |L - x_n| + |x_n - M|$$

$$= |x_n - L| + |x_n - M|$$

$$< \varepsilon + \varepsilon$$

$$= 2\varepsilon = |L - M|$$

ie, For $n \geq N$ $\boxed{|L - M| < |L - M|}$

This is a contradiction.

So we conclude that in fact $L = M$, as required.